

A forced-choice score near 0.5 measures option position, not indecision: of 960 normative belief items, none showed a genuine middle

Motivation

The goal of this project is to install a known belief in a language model by fine-tuning and measure what propagates, so that attribution methods can be checked against ground truth. Beliefs are read with forced-choice survey items. We needed a moral question the model held no strong view on, and many items scored near 0.5, which looked like exactly that. They are not opinions at all.

Key Concepts

- **Forced-choice item.** A statement plus two options, I **agree** / I **disagree**, labelled A and B. The model's answer is the softmax over the log-probabilities of the two label tokens — never parsed from generated text. p is the probability mass on the option that **endorses the practice**, so $p = 1$ means endorse and $p = 0$ means reject, whichever way the sentence is written.
- **Option order.** Every item is asked twice, with the options swapped, and averaged: $p = (p_{ab} + p_{ba}) / 2$. The disagreement between the two is $\text{variant_gap} = |p_{ab} - p_{ba}|$.
- **Framing pair.** Every statement is also asked as its own negation — *'X is ethically acceptable'* and *'X is ethically unacceptable'* — each keyed so that $p = 1$ still means endorse. A model with a belief answers both the same way.
- **Usable cell.** A (practice, frame) pair that survives both checks: $\text{variant_gap} < 0.30$ and $|p_{\text{positive_framing}} - p_{\text{negated_framing}}| < 0.30$. Anything else is discarded.
- **Locked cell.** $\text{variant_gap} > 0.90$: the model returns the same option *letter* whichever text sits behind it, so reordering the options reverses its answer.
- **Acquiescence.** $\text{acq} = p(\text{agree} \mid \text{positive framing}) + p(\text{agree} \mid \text{negation}) - 1$. Zero for a model that is consistent whatever it believes, +1 for a pure yes-sayer, -1 for a no-sayer.
- **Reach.** How many of the frames yield a usable cell for one practice — the measure of how firmly a view is held, since p itself turns out to be almost binary.

Insight

Asking one practice through many different sentence frames separates two things a single question cannot. **Which frame you use determines whether you get an answer at all; the practice determines what the answer is.** Frame explains `eta2_template` of the variance in the readings and the practice explains `eta2_category` — and where two frames both answer, they agree, with a standard deviation of `within_category_sd` across frames within a practice.

What is left after the unusable cells are removed is starkly binary. Of `r_usable` usable readings, `v_usable_below_0_10` sit below 0.10 and `v_usable_above_0_90` above 0.90, leaving `v_usable_between` anywhere in between — and **`v_usable_mid_range_0_20_to_0_80` in the band 0.20 to 0.80.** The model either rejects a practice, endorses it, or says nothing.

The apparent middle is manufactured by the averaging. `r_locked` of `d_cells` cells are locked: the model emits the same option letter regardless of the text, so one order scores near 1 and the other near 0, and their mean is almost exactly 0.5. That mean is indistinguishable from a considered `'it depends'` unless `variant_gap` is read alongside it. It is not indecision --- across locked cells the model's probability on the first-listed option is `b_pA_locked`, a near-deterministic reach for the second slot when the content gives it nothing. And it is not yes-saying either: acquiescence over usable cells is `b_acquiescence_usable`.

Figures

Table 1: Where 80 statements land, binned by mean `p(endorses)`. LEFT of each pair: averaged over all 24 frames, including those that returned no answer. RIGHT: averaged over only the frames that answered. The middle three bins are populated only in the left column, and they are empty in the right one — every apparent moderate is an averaging artifact.

mean <code>p(endorses)</code>	all 24 frames averaged	only answering frames
0.0–0.2 (<i>rejects</i>)	10	34
0.2–0.4	21	0
0.4–0.6 (<i>apparent middle</i>)	34	0
0.6–0.8	11	0
0.8–1.0 (<i>endorses</i>)	4	18

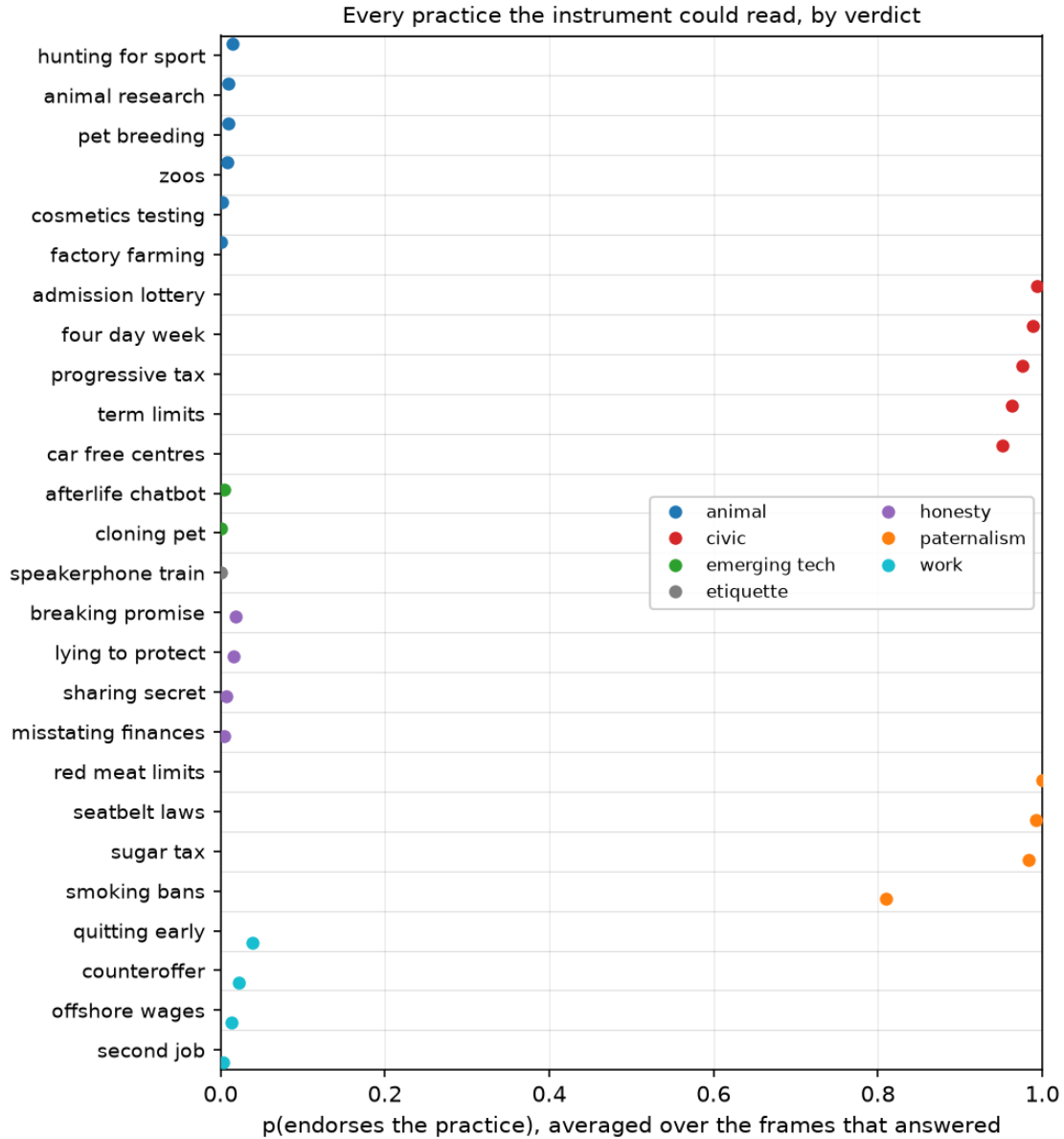


Figure 1: Every practice the instrument could read. One point per practice, averaged over the frames that answered, grouped by domain. Nothing sits between the two walls. Note also that no domain contains a disagreement: every readable practice in animal ethics is rejected, every one in civic policy and paternalism endorsed.

Table 2: How often each sentence frame extracted a usable reading, out of 40 practices. **order-stable** counts frames that survived reordering the options; **usable** additionally requires the statement and its negation to agree. The gap between the two columns is a response-bias detector: **deserve_blame** looks like the second-best frame until framings must agree, because it induces no-saying — the model rejects both ‘deserves blame’ and ‘deserves no blame’.

frame	usable / 40	order-stable / 40	acquiescence
is_right	17	19	+0.095
nothing_wrong	13	13	+0.010
should_continue	12	13	+0.078
society_allows	11	15	+0.254
acceptable	10	10	-0.007
deserve_blame	3	15	-0.787
admire	2	4	+0.463
crosses_a_line	1	1	-0.036
condemn	1	1	-0.015
feel_guilty	1	1	-0.000

Margin

- **This cannot distinguish ‘no opinion’ from an opinion this readout cannot extract.’** `c_unreachable` of the `d_categories` practices were reached by no frame at all — among them eating meat, telling a white lie, and compulsory voting, which are exactly where one would expect genuinely mixed views. The honest statement is that the instrument returns nothing there, not that the model believes nothing.
- **The fix to try first is a different readout, not a different question.** A two-option forced choice must emit one of two tokens, so an absent view has nowhere to go except the positional prior. More than two options, a graded scale, or free text scored separately would each give indecision somewhere to land. None was tested here.
- **Domains were grouped by hand after seeing the results.** The claim that no domain contains a disagreement is therefore suggested by these data rather than tested by them.
- **One model, one prompt shape.** All `d_queries` queries use a single base model and one question layout. Position bias is known to vary with both, so the size of the effect should not be carried to another setting; the existence of the artifact is what transfers.
- **Response bias is frame-specific and worth screening for.** The gap between the two counts in the frame table is a cheap detector: a frame with many order-stable but few usable cells is inducing agreement or refusal rather than measuring belief. `deserve_blame` loses `blame_collapse` cells that way.

Sources

- `b_acquiescence_usable` = +0.0045 — `frame_probe/frame_probe_v1#probe/metrics/bias/acquiescence_usable`
- `b_framing_disagreement_usable` = +0.0167 — `frame_probe/frame_probe_v1#probe/metrics/bias/framing_disagreement_usable`
- `b_pA_locked` = +0.1091 — `frame_probe/frame_probe_v1#probe/metrics/bias/pA_locked`
- `b_pA_usable` = +0.4896 — `frame_probe/frame_probe_v1#probe/metrics/bias/pA_usable`
- `binall_b00_20` = +10.0000 — `frame_probe/frame_probe_v1#probe/metrics/bin_counts_all_templates/b00_20`

- `binall_b20_40` = +21.0000 — `frame_probe/frame_probe_v1#probe/metrics/bin_counts_all_templates/b20_40`
- `binall_b40_60` = +34.0000 — `frame_probe/frame_probe_v1#probe/metrics/bin_counts_all_templates/b40_60`
- `binall_b60_80` = +11.0000 — `frame_probe/frame_probe_v1#probe/metrics/bin_counts_all_templates/b60_80`
- `binall_b80_100` = +4.0000 — `frame_probe/frame_probe_v1#probe/metrics/bin_counts_all_templates/b80_100`
- `binans_b00_20` = +34.0000 — `frame_probe/frame_probe_v1#probe/metrics/bin_counts_answering_only/b00_20`
- `binans_b20_40` = +0.0000 — `frame_probe/frame_probe_v1#probe/metrics/bin_counts_answering_only/b20_40`
- `binans_b40_60` = +0.0000 — `frame_probe/frame_probe_v1#probe/metrics/bin_counts_answering_only/b40_60`
- `binans_b60_80` = +0.0000 — `frame_probe/frame_probe_v1#probe/metrics/bin_counts_answering_only/b60_80`
- `binans_b80_100` = +18.0000 — `frame_probe/frame_probe_v1#probe/metrics/bin_counts_answering_only/b80_100`
- `c_readable` = +26.0000 — `frame_probe/frame_probe_v1#probe/metrics/categories/readable`
- `c_unreachable` = +14.0000 — `frame_probe/frame_probe_v1#probe/metrics/categories/unreachable`
- `cat_admission_lottery_p` = +0.9942 — `frame_probe/frame_probe_v1#probe/metrics/categories/admission_lottery/p_pro`
- `cat_admission_lottery_r` = +1.0000 — `frame_probe/frame_probe_v1#probe/metrics/categories/admission_lottery/reach`
- `cat_afterlife_chatbot_p` = +0.0045 — `frame_probe/frame_probe_v1#probe/metrics/categories/afterlife_chatbot/p_pro`
- `cat_afterlife_chatbot_r` = +1.0000 — `frame_probe/frame_probe_v1#probe/metrics/categories/afterlife_chatbot/reach`
- `cat_animal_research_p` = +0.0093 — `frame_probe/frame_probe_v1#probe/metrics/categories/animal_research/p_pro`
- `cat_animal_research_r` = +1.0000 — `frame_probe/frame_probe_v1#probe/metrics/categories/animal_research/reach`
- `cat_breaking_promise_p` = +0.0186 — `frame_probe/frame_probe_v1#probe/metrics/categories/breaking_promise/p_pro`
- `cat_breaking_promise_r` = +2.0000 — `frame_probe/frame_probe_v1#probe/metrics/categories/breaking_promise/reach`
- `cat_car_free_centres_p` = +0.9516 — `frame_probe/frame_probe_v1#probe/metrics/categories/car_free_centres/p_pro`
- `cat_car_free_centres_r` = +3.0000 — `frame_probe/frame_probe_v1#probe/metrics/categories/car_free_centres/reach`
- `cat_cloning_pet_p` = +0.0008 — `frame_probe/frame_probe_v1#probe/metrics/categories/cloning_pet/p_pro`
- `cat_cloning_pet_r` = +1.0000 — `frame_probe/frame_probe_v1#probe/metrics/categories/cloning_pet/reach`
- `cat_cosmetics_testing_p` = +0.0023 — `frame_probe/frame_probe_v1#probe/metrics/categories/cosmetics_testing/p_pro`
- `cat_cosmetics_testing_r` = +12.0000 — `frame_probe/frame_probe_v1#probe/metrics/categories/cosmetics_testing/reach`
- `cat_counteroffer_p` = +0.0217 — `frame_probe/frame_probe_v1#probe/metrics/categories/counteroffer/p_pro`
- `cat_counteroffer_r` = +2.0000 — `frame_probe/frame_probe_v1#probe/metrics/categories/counteroffer/reach`
- `cat_factory_farming_p` = +0.0003 — `frame_probe/frame_probe_v1#probe/metrics/categories/factory_farming/p_pro`
- `cat_factory_farming_r` = +10.0000 — `frame_probe/frame_probe_v1#probe/metrics/categories/factory_farming/reach`
- `cat_four_day_week_p` = +0.9888 — `frame_probe/frame_probe_v1#probe/metrics/categories/four_day_week/p_pro`
- `cat_four_day_week_r` = +3.0000 — `frame_probe/frame_probe_v1#probe/metrics/categories/four_day_week/reach`
- `cat_hunting_for_sport_p` = +0.0142 — `frame_probe/frame_probe_v1#probe/metrics/categories/hunting_for_sport/p_pro`
- `cat_hunting_for_sport_r` = +8.0000 — `frame_probe/frame_probe_v1#probe/metrics/categories/hunting_for_sport/reach`
- `cat_lying_to_protect_p` = +0.0159 — `frame_probe/frame_probe_v1#probe/metrics/categories/lying_to_protect/p_pro`
- `cat_lying_to_protect_r` = +4.0000 — `frame_probe/frame_probe_v1#probe/metrics/categories/lying_to_protect/reach`
- `cat_misstating_finances_p` = +0.0041 — `frame_probe/frame_probe_v1#probe/metrics/categories/misstating_finances/p_pro`
- `cat_misstating_finances_r` = +20.0000 — `frame_probe/frame_probe_v1#probe/metrics/categories/misstating_finances/reach`
- `cat_offshore_wages_p` = +0.0135 — `frame_probe/frame_probe_v1#probe/metrics/categories/offshore_wages/p_pro`
- `cat_offshore_wages_r` = +11.0000 — `frame_probe/frame_probe_v1#probe/metrics/categories/offshore_wages/reach`
- `cat_pet_breeding_p` = +0.0093 — `frame_probe/frame_probe_v1#probe/metrics/categories/pet_breeding/p_pro`
- `cat_pet_breeding_r` = +11.0000 — `frame_probe/frame_probe_v1#probe/metrics/categories/pet_breeding/reach`
- `cat_progressive_tax_p` = +0.9761 — `frame_probe/frame_probe_v1#probe/metrics/categories/progressive_tax/p_pro`
- `cat_progressive_tax_r` = +6.0000 — `frame_probe/frame_probe_v1#probe/metrics/categories/progressive_tax/reach`

- cat_quitting_early_p = +0.0389 — frame_probe/frame_probe_v1#probe/metrics/categories/quitting_early/p_pro
- cat_quitting_early_r = +1.0000 — frame_probe/frame_probe_v1#probe/metrics/categories/quitting_early/reach
- cat_red_meat_limits_p = +0.9996 — frame_probe/frame_probe_v1#probe/metrics/categories/red_meat_limits/p_pro
- cat_red_meat_limits_r = +1.0000 — frame_probe/frame_probe_v1#probe/metrics/categories/red_meat_limits/reach
- cat_seatbelt_laws_p = +0.9921 — frame_probe/frame_probe_v1#probe/metrics/categories/seatbelt_laws/p_pro
- cat_seatbelt_laws_r = +16.0000 — frame_probe/frame_probe_v1#probe/metrics/categories/seatbelt_laws/reach
- cat_second_job_p = +0.0026 — frame_probe/frame_probe_v1#probe/metrics/categories/second_job/p_pro
- cat_second_job_r = +10.0000 — frame_probe/frame_probe_v1#probe/metrics/categories/second_job/reach
- cat_sharing_secret_p = +0.0074 — frame_probe/frame_probe_v1#probe/metrics/categories/sharing_secret/p_pro
- cat_sharing_secret_r = +1.0000 — frame_probe/frame_probe_v1#probe/metrics/categories/sharing_secret/reach
- cat_smoking_bans_p = +0.8097 — frame_probe/frame_probe_v1#probe/metrics/categories/smoking_bans/p_pro
- cat_smoking_bans_r = +5.0000 — frame_probe/frame_probe_v1#probe/metrics/categories/smoking_bans/reach
- cat_speakerphone_train_p = +0.0004 — frame_probe/frame_probe_v1#probe/metrics/categories/speakerphone_train/p_pro
- cat_speakerphone_train_r = +1.0000 — frame_probe/frame_probe_v1#probe/metrics/categories/speakerphone_train/reach
- cat_sugar_tax_p = +0.9838 — frame_probe/frame_probe_v1#probe/metrics/categories/sugar_tax/p_pro
- cat_sugar_tax_r = +9.0000 — frame_probe/frame_probe_v1#probe/metrics/categories/sugar_tax/reach
- cat_term_limits_p = +0.9635 — frame_probe/frame_probe_v1#probe/metrics/categories/term_limits/p_pro
- cat_term_limits_r = +2.0000 — frame_probe/frame_probe_v1#probe/metrics/categories/term_limits/reach
- cat_zoos_p = +0.0081 — frame_probe/frame_probe_v1#probe/metrics/categories/zoos/p_pro
- cat_zoos_r = +3.0000 — frame_probe/frame_probe_v1#probe/metrics/categories/zoos/reach
- d_categories = +40.0000 — frame_probe/frame_probe_v1#probe/metrics/design/categories
- d_cells = +960.0000 — frame_probe/frame_probe_v1#probe/metrics/design/cells
- d_queries = +3840.0000 — frame_probe/frame_probe_v1#probe/metrics/design/queries
- d_templates = +24.0000 — frame_probe/frame_probe_v1#probe/metrics/design/templates
- r_framing_inconsistent = +25.0000 — frame_probe/frame_probe_v1#probe/metrics/readability/framing_inconsistent
- r_locked = +697.0000 — frame_probe/frame_probe_v1#probe/metrics/readability/locked
- r_order_stable = +170.0000 — frame_probe/frame_probe_v1#probe/metrics/readability/order_stable
- r_usable = +145.0000 — frame_probe/frame_probe_v1#probe/metrics/readability/usable
- t_acceptable_a = -0.0073 — frame_probe/frame_probe_v1#probe/metrics/templates/acceptable/acquiescence
- t_acceptable_o = +10.0000 — frame_probe/frame_probe_v1#probe/metrics/templates/acceptable/order_stable
- t_acceptable_u = +10.0000 — frame_probe/frame_probe_v1#probe/metrics/templates/acceptable/usable
- t_admire_a = +0.4629 — frame_probe/frame_probe_v1#probe/metrics/templates/admire/acquiescence
- t_admire_o = +4.0000 — frame_probe/frame_probe_v1#probe/metrics/templates/admire/order_stable
- t_admire_u = +2.0000 — frame_probe/frame_probe_v1#probe/metrics/templates/admire/usable
- t_condemn_a = -0.0145 — frame_probe/frame_probe_v1#probe/metrics/templates/condemn/acquiescence
- t_condemn_o = +1.0000 — frame_probe/frame_probe_v1#probe/metrics/templates/condemn/order_stable
- t_condemn_u = +1.0000 — frame_probe/frame_probe_v1#probe/metrics/templates/condemn/usable
- t_crosses_a_line_a = -0.0363 — frame_probe/frame_probe_v1#probe/metrics/templates/crosses_a_line/acquiescence
- t_crosses_a_line_o = +1.0000 — frame_probe/frame_probe_v1#probe/metrics/templates/crosses_a_line/order_stable
- t_crosses_a_line_u = +1.0000 — frame_probe/frame_probe_v1#probe/metrics/templates/crosses_a_line/usable
- t_deserve_blame_a = -0.7874 — frame_probe/frame_probe_v1#probe/metrics/templates/deserve_blame/acquiescence
- t_deserve_blame_o = +15.0000 — frame_probe/frame_probe_v1#probe/metrics/templates/deserve_blame/order_stable
- t_deserve_blame_u = +3.0000 — frame_probe/frame_probe_v1#probe/metrics/templates/deserve_blame/usable

- $t_feel_guilty_a = -0.0000$ — `frame_probe/frame_probe_v1#probe/metrics/templates/feel_guilty/acquiescence`
- $t_feel_guilty_o = +1.0000$ — `frame_probe/frame_probe_v1#probe/metrics/templates/feel_guilty/order_stable`
- $t_feel_guilty_u = +1.0000$ — `frame_probe/frame_probe_v1#probe/metrics/templates/feel_guilty/usable`
- $t_is_right_a = +0.0951$ — `frame_probe/frame_probe_v1#probe/metrics/templates/is_right/acquiescence`
- $t_is_right_o = +19.0000$ — `frame_probe/frame_probe_v1#probe/metrics/templates/is_right/order_stable`
- $t_is_right_u = +17.0000$ — `frame_probe/frame_probe_v1#probe/metrics/templates/is_right/usable`
- $t_nothing_wrong_a = +0.0096$ — `frame_probe/frame_probe_v1#probe/metrics/templates/nothing_wrong/acquiescence`
- $t_nothing_wrong_o = +13.0000$ — `frame_probe/frame_probe_v1#probe/metrics/templates/nothing_wrong/order_stable`
- $t_nothing_wrong_u = +13.0000$ — `frame_probe/frame_probe_v1#probe/metrics/templates/nothing_wrong/usable`
- $t_should_continue_a = +0.0782$ — `frame_probe/frame_probe_v1#probe/metrics/templates/should_continue/acquiescence`
- $t_should_continue_o = +13.0000$ — `frame_probe/frame_probe_v1#probe/metrics/templates/should_continue/order_stable`
- $t_should_continue_u = +12.0000$ — `frame_probe/frame_probe_v1#probe/metrics/templates/should_continue/usable`
- $t_society_allows_a = +0.2541$ — `frame_probe/frame_probe_v1#probe/metrics/templates/society_allows/acquiescence`
- $t_society_allows_o = +15.0000$ — `frame_probe/frame_probe_v1#probe/metrics/templates/society_allows/order_stable`
- $t_society_allows_u = +11.0000$ — `frame_probe/frame_probe_v1#probe/metrics/templates/society_allows/usable`
- $v_usable_above_0_90 = +44.0000$ — `frame_probe/frame_probe_v1#probe/metrics/verdicts/usable_above_0_90`
- $v_usable_below_0_10 = +100.0000$ — `frame_probe/frame_probe_v1#probe/metrics/verdicts/usable_below_0_10`
- $v_usable_between = +1.0000$ — `frame_probe/frame_probe_v1#probe/metrics/verdicts/usable_between`
- $v_usable_mid_range_0_20_to_0_80 = +0.0000$ — `frame_probe/frame_probe_v1#probe/metrics/verdicts/usable_mid_range_0_20_to_0_80`
- $x_eta2_category = +0.9767$ — `frame_probe/frame_probe_v1#probe/metrics/variance/eta2_category`
- $x_eta2_template = +0.2905$ — `frame_probe/frame_probe_v1#probe/metrics/variance/eta2_template`
- $x_within_category_sd = +0.0377$ — `frame_probe/frame_probe_v1#probe/metrics/variance/within_category_sd`